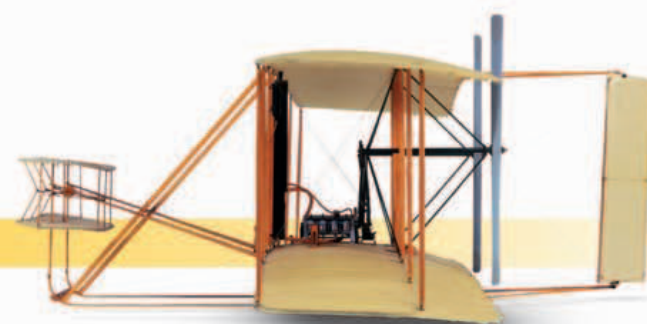


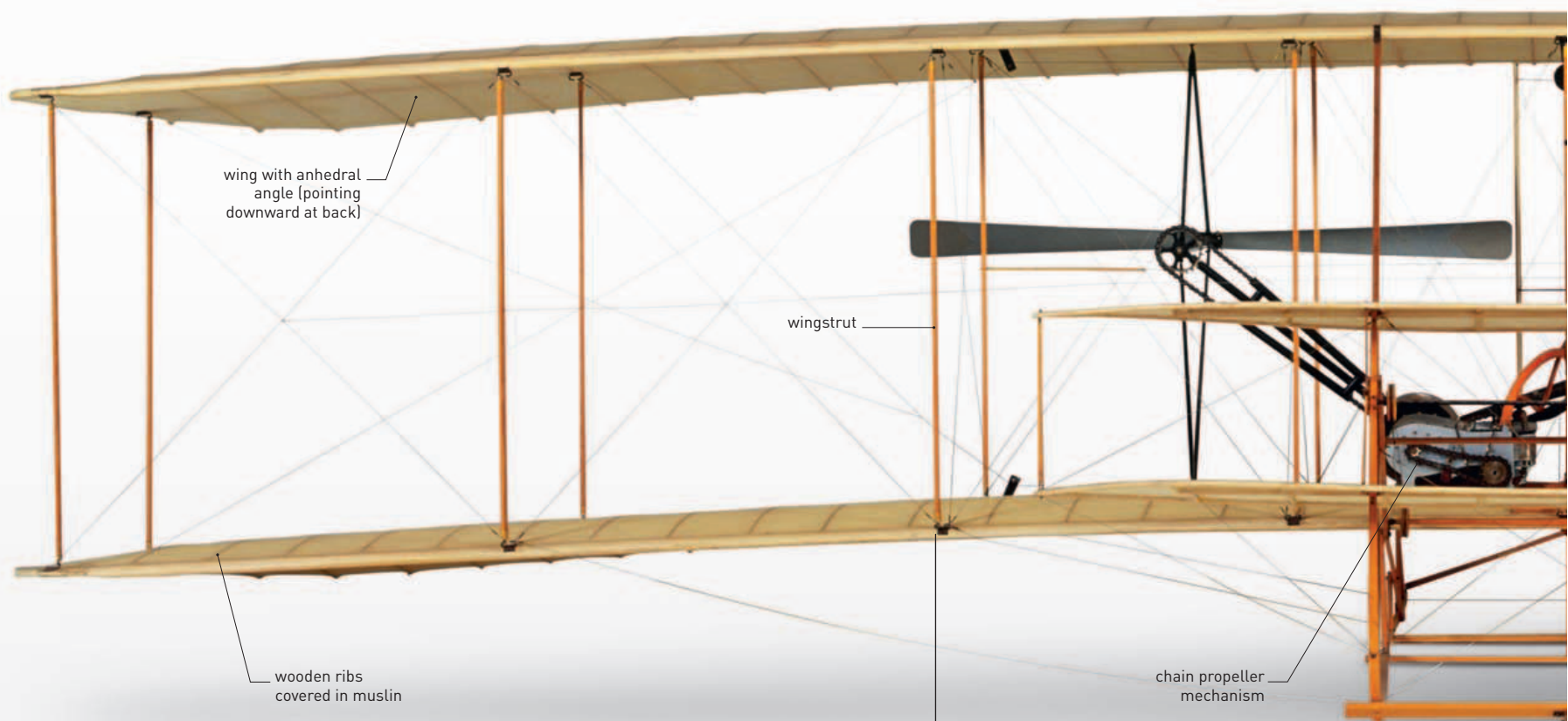
THE STORY OF FLIGHT

FROM HOT-AIR BALLOONS TO SUPERSONIC JETS, FLIGHT HAS COME A LONG WAY

Until the 20th century, human flight was an area of experiment for enthusiasts. Ascents in balloons sparked the first flying craze but had little practical effect. In the 19th century, engineers calculated the forces involved in winged flight and experimented with gliders, but it was the arrival of gasoline engines that made powered flight practicable.



Side view



1783

Hot-air balloon

French brothers Joseph and Etienne Montgolfier complete the first manned flight in a hot-air balloon.



Montgolfier balloon

Zeppelin airship

1900

First Zeppelin flight

On July 2, German pioneer Ferdinand von Zeppelin's LZ-1 successfully takes to the skies.



1903

The Wright Flyer

On December 17, the Wright brothers complete the first sustained, controlled flight in a powered, heavier-than-air machine at Kill Devil Hills, North Carolina, US.

1919

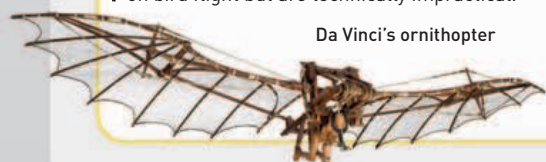
Airlines progress

The first scheduled international passenger air service is inaugurated between London and Paris; the first airlines are set up.

c. 1485-1510

Leonardo's flying machine

Early concepts of human flight, like those sketched by Leonardo da Vinci, are based on bird flight but are technically impractical.

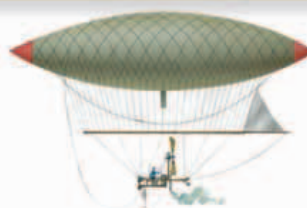


Da Vinci's ornithopter

1852

First powered flight

Frenchman Henri Giffard attaches a steam engine to a balloon filled with coal gas; powered flight begins.



Giffard's airship

1909

Long-distance flight

On July 25, Frenchman Louis Blériot flies across the English Channel from France to England.



Louis Blériot

1914

Aircraft in warfare

Use of aircraft for combat transforms aviation; tens of thousands of aircraft are mass-produced for the first time.